

Medial-axis reconstruction on complete Riemannian manifolds: capture, drift, and the medial complex*

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Abstract

Let X be a nonempty proper closed subset of a complete Riemannian manifold M . The *two-geodesic medial axis* M_X^g is the set of points admitting at least two minimising geodesics to X — the multi-geodesic part of the cut locus of X — and the associated *reconstruction* $\mathcal{R}_X^g$ is the union of the open free balls $B(a, d_X(a))$ over $a \in M_X^g$. We characterise $\mathcal{R}_X^g$ on every complete M , with no curvature, genericity, spread, or cardinality hypothesis. A *capture theorem* shows that along any solution of the differential inclusion $\Phi' \in -\text{conv } U(\Phi)$, driven by the unit directions of minimising geodesics, the distance d_X grows at unit rate until the first approach to M_X^g ; hence every point either lies at depth arbitrarily close to $d_X(p)$ inside a medial ball, or escapes to infinity. In particular $\mathcal{R}_X^g = M \setminus X$ on every compact manifold, for every closed X — even a single point — answering a question of Białożyty. Normalising free balls by their reach, the locally uniform limits $\psi = \lim[d(\cdot, a_k) - d_X(a_k)]$ over medial centres — *medial limit functions* — satisfy $\{\psi < 0\} \subseteq \mathcal{R}_X^g$, and $\mathcal{R}_X^g$ is exactly the union of these sublevel sets; every point is captured, drift-swallowed, or escaping, and Białożyty’s Euclidean theorem is the flat instance. A topological layer follows: the escape class coincides with the Aubry set of d_X in the sense of Cannarsa–Cheng–Fathi, so the captured part of the complement always has the homotopy type of M_X^g , and $M \setminus X \simeq M_X^g \simeq \text{Cut}(X)$ whenever $\sup d_X < \infty$. Finally, every swallowed-uncaptured point lies in an *ideal* free horoball, so the *medial complex* — M_X^g with the ideal free horoballs glued along the pattern of their overlaps — always carries the homotopy type of $\mathcal{R}_X^g$.

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*Drafted with the assistance of Claude (Anthropic); all arguments were re-derived and checked by hand, and the two mechanisms of the capture theorem — the rate-one identity off the axis and the margin at first approach — were verified numerically on $\mathbb{S}^2$ against a non-circular convex contact curve whose focal region forces the foot to slide (60 random starts: growth rate 1 to five digits, margin deficit $< 3 \cdot 10^{-6}$; ancillary file `medial_capture_check.py`). An earlier version of parts of this material circulated as a series of informal notes. Remaining errors are ours.

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1 Introduction

1.1 The reconstruction problem

Fix a nonempty closed set X in a metric space M and, for each point p outside X , ask for the nearest points of X . Where the answer is unique the geometry of X is locally transparent; the interesting locus is the set of p with an *ambiguous* nearest-point answer. In the plane this is the classical *medial axis* or *skeleton*, a one-dimensional graph that threads the region between the features of X and has been used, since Blum, as a shape descriptor in vision, in mesh generation, and in the topology of domains [7]. Two features of the axis make it a genuine *descriptor* rather than a mere by-product: each axis point a carries a scalar, the distance $d_X(a)$ to X , and the open ball $B(a, d_X(a))$ of that radius is a *free ball* — a maximal ball that meets X only in its boundary. The union of the free balls over the axis,

$$\mathcal{R}_X = \bigcup_{a \in M_X} B(a, d_X(a)),$$

is the *reconstruction* of the complement of X from the axis.

One wants to know when the reconstruction is faithful, i.e. when $\mathcal{R}_X = M \setminus X$. The obstruction is easy to see and hard to characterise. If X is convex in $\mathbb{R}^n$ the axis is empty and $\mathcal{R}_X = \emptyset$: convex sets cast no medial shadow, so their complements are not reconstructed at all. If X is a pair of points in the plane, the axis is the perpendicular bisector and its free balls reconstruct almost everything: the central ball already swallows the open segment between the points, and balls centred ever farther out along the bisector fill the two open half-planes bounded by the line through the pair — the deeper the point, the more remote the ball that reaches it, and the organising object is the *limit* of the escaping balls, an open half-plane. What no ball ever reaches are the two collinear rays beyond the points. The interplay between the local balls, which fill the “inside” of the hull, and the escaping balls, whose limits fill the outer half-spaces of defective faces of the hull, is the whole content of the Euclidean theorem.

1.2 Białożył's theorem, the starting point

Białożył [4] settled the Euclidean case completely. Write $\text{conv } X$ for the convex hull and $\overline{\text{conv}} X$ for its closure. A supporting hyperplane L of $\overline{\text{conv}} X$ is *defective* if $X \cap L \neq \overline{\text{conv}} X \cap L$, i.e. if X fails to fill the contact face that the hull presents on L ; for each supporting $L = \{\langle v, \cdot \rangle = c\}$ (with $X \subseteq \{\langle v, \cdot \rangle \leq c\}$) let $L^+ = \{\langle v, \cdot \rangle > c\}$ be the open outer half-space.

Theorem 1.1 (Białożył [4, Thm. 6]). *Let $X \subseteq \mathbb{R}^n$ be closed and nonempty. Then X is reconstructible, i.e. $\mathcal{R}_X = \mathbb{R}^n \setminus X$, if and only if*

$$\mathbb{R}^n \setminus X \subseteq \overline{\text{conv}} X \cup \bigcup_{L \text{ defective}} L^+.$$

Two features of this statement organise everything that follows. First, the hull $\overline{\text{conv}} X$ is reconstructed by *ordinary* (finite) medial balls: every point of $\overline{\text{conv}} X \setminus X$ lies in a free ball centred on the axis [4]. Second, the outer half-spaces L^+ of defective supporting hyperplanes are reconstructed only in the *limit*, by free balls whose centres escape to infinity along a direction normal to L ; a supporting hyperplane contributes such a limit precisely when its contact set is defective — equivalently, by Motzkin's theorem [10], when a point of L just outside the contact face is non-Chebyshev. This is the dichotomy *local balls vs. escaping balls* in its cleanest form, and the first goal of this paper is to find its counterpart on a curved manifold, where “half-space” must become “horoball”, “convex hull” must become something curvature-dependent, and the very notion of “two nearest points” must be re-examined because a single nearest point can be reached by several competing geodesics. The second goal is the question Białożył's theorem raises but cannot ask in $\mathbb{R}^n$: what happens on a *compact* manifold, where there is no infinity for the centres to escape to — a question posed explicitly in [4, §5.2]. The third goal, occupying the second half of the paper, is topological: when, and in what form, the axis carries the *homotopy type* of what it reconstructs.

1.3 From $\mathbb{R}^n$ to a manifold: what breaks

Let M be a complete Riemannian manifold. Three Euclidean conveniences fail at once.

(i) *Geodesics are not unique.* Two points can be joined by several minimising geodesics, and the nearest-point map to X can be single-valued (a unique foot $y \in X$) yet ambiguous as a *geodesic* problem (several minimisers reaching that same y). The sphere makes this vivid: for X a single point, every other point has that point as its unique nearest point, yet from the antipode it is reached by a whole sphere's worth of minimising geodesics — one nearest point, many nearest geodesics. This forces a distinction, invisible in $\mathbb{R}^n$, between the *two-point* axis M_X (two nearest *points*) and the *two-geodesic* axis M_X^g (two minimising *geodesics*); $M_X \subseteq M_X^g$, and it is M_X^g — the multi-geodesic part of the cut locus of X — that carries the reconstruction theory.

(ii) *Convexity is not linear.* There is no “ $\text{conv } X$ ”; its role is played by different objects in different curvatures — the ordinary hull only in the flat case, a horoconvex hull in nonpositive curvature, and *nothing at all* on a compact manifold, where every complement turns out to be reconstructed and the hull condition is vacuous.

(iii) *The distance function turns.* In $\mathbb{R}^n$ the ascending gradient flow of d_X runs along straight rays until it meets the bisector; on a manifold the minimising geodesic to X bends, its foot slides, and the natural “margin” one would monotonically accumulate along the flow genuinely oscillates

as the active contact set changes. This is the analytic heart of the difficulty and the reason a naive gradient-flow argument stalls exactly at the cut points, where it is asked to continue past a singularity of d_X . A further, quieter difficulty compounds it: M_X^g is *not closed* — minimising geodesics can merge in the limit at focal degenerations — so “the first time the flow meets the axis” need not exist.

The resolution, developed below, is to change the question the flow is asked to answer. Rather than flowing *through* the axis and tracking a monotone quantity, we flow *up to* the axis and read off a margin that is exact on the complement of the axis and merely needs to survive to the *first approach* — a formulation indifferent to whether the axis is closed and to everything that happens at focal and conjugate points. This severs the argument from every regularity subtlety at cut points, and it is what makes the theory unconditional.

1.4 Results

The paper proves, on every complete Riemannian manifold M and for every nonempty closed proper X , three groups of results.

Capture (§3). Along any trajectory of the ascent inclusion $\Phi' \in -\operatorname{conv} U(\Phi)$, and up to the first approach to M_X^g , the distance d_X grows at rate exactly one while the trajectory is 1-Lipschitz (Theorem 3.1); so each point either sits at depth close to $d_X(p)$ inside a moving medial ball — *capture* — or *escapes* to infinity. Hence $\mathcal{R}_X^g = M \setminus X$ on *every* compact manifold, for every closed X , even a single point (Corollary 3.2), answering the compact reconstruction question of [4, §5.2]; a spread refinement gives the two-point statement under $\sup d_X < \operatorname{inj}(M)$ (Corollary 3.3).

Drift and the characterisation (§4). Normalising free balls by their reach, the locally uniform limits $\psi = \lim[d(\cdot, a_k) - d_X(a_k)]$ over medial centres — *medial limit functions* (Definition 2.3) — each reconstruct their whole open sublevel set (Lemma 4.1), and $\mathcal{R}_X^g$ is exactly the union of these sublevel sets (Theorem 4.3), yielding the trichotomy *captured* / *drift-swallowed* / *escaping* (Theorem 4.4) and a reconstructibility criterion (Corollary 4.5) of which Białyżyt’s Theorem 1.1 is the flat instance (Corollary 4.6).

The topological layer (§5–§6). The escape class is exactly the Aubry set of d_X in the sense of Cannarsa–Cheng–Fathi [5], so the captured part of the complement always has the homotopy type of M_X^g (Theorem 5.5), and $M \setminus X \simeq M_X^g \simeq \operatorname{Cut}(X)$ whenever $\sup d_X < \infty$ (Corollary 5.3). Swallowing, metrically complete as it may be, does not transport homotopy type; the medial complex repairs this — every swallowed-uncaptured point lies in an *ideal* free horoball (Theorem 6.3), so M_X^g with the ideal free horoballs glued on carries the homotopy type of $\mathcal{R}_X^g$ (Theorem 6.5).

1.5 Related work

The distance function from a closed set, its singular set, and its gradient dynamics have a substantial literature. What is new here: the capture theorem extracts a quantitative reconstruction margin, for an arbitrary closed set in an arbitrary complete manifold, by *stopping* at the (non-closed) axis where the singular gradient-flow theorems of Lieutier [7] and Albano–Cannarsa–Nguyen–Sinestrari [2] are designed to continue; the medial limit functions make the escaping balls a theorem-free limit object on any complete manifold; and the homotopy results localise the failure of topological faithfulness in the Aubry set of Cannarsa–Cheng–Fathi [5] and repair it with the ideal cells of the medial complex. The fine structure theory of the cut locus of

a closed set (Tanaka–Sabau [11], Miura–Tanaka [9]) is not needed for the arguments but is the natural tool for the regularity questions the complex raises.

2 Preliminaries

Throughout, M is a complete Riemannian manifold; by the Hopf–Rinow theorem it is a proper geodesic metric space (closed balls are compact, and any two points are joined by a minimising geodesic). $X \subseteq M$ is closed with $\emptyset \neq X \neq M$, and $d_X = \text{dist}(\cdot, X)$ is the distance to X , a 1-Lipschitz function, positive exactly off X . All geodesics are parametrised by arc length. We state the standing hypotheses once, to prevent a misreading the subject invites: *complete metric, closed X — nothing else*. No curvature assumption of any kind (sign, constancy, pinching, bounds) appears in §2–§4; the curvature of M may vary and change sign freely.

2.1 The two axes and the two reconstructions

For $x \in M \setminus X$ a *minimising geodesic to X* is a unit-speed geodesic $\gamma : [0, d_X(x)] \rightarrow M$ with $\gamma(0) = x$ and $\gamma(d_X(x)) \in X$; its *foot* is $\gamma(d_X(x))$. Set

$$U(x) = \{\gamma'(0) : \gamma \text{ a minimising geodesic to } X, \gamma(0) = x\} \subseteq T_x^1 M,$$

the set of unit initial directions of such geodesics (a geodesic is determined by its initial vector, so distinct minimisers have distinct directions). Define the two medial axes

$$M_X = \{a : a \text{ has } \geq 2 \text{ nearest points in } X\}, \quad M_X^g = \{x \in M \setminus X : |U(x)| \geq 2\},$$

the *two-point* and *two-geodesic* axes. Because two distinct minimising geodesics may share a foot, $M_X \subseteq M_X^g$; and M_X^g is the multi-geodesic part of the cut locus of X — in general a proper dense subset of it, the cut locus also containing focal points reached by a unique minimiser. Their reconstructions are

$$\mathcal{R}_X = \bigcup_{a \in M_X} B(a, d_X(a)), \quad \mathcal{R}_X^g = \bigcup_{a \in M_X^g} B(a, d_X(a)), \quad \mathcal{R}_X \subseteq \mathcal{R}_X^g,$$

unions of *open* free balls. When we use M_X^g as a target to be reached we write $\Sigma := M_X^g$; $r(a) = d_X(a)$ is the *reach* of a centre a . In $\mathbb{R}^n$ geodesics between points are unique, so $M_X = M_X^g$ and $\mathcal{R}_X = \mathcal{R}_X^g$, and the two theories coincide; the split is a purely curved phenomenon.

The two-geodesic axis is the correct object: a single point $X = \{x_0\}$ on the round sphere has $M_X = \emptyset$ — every point off x_0 has the unique nearest point x_0 — yet the antipode $\bar{x}_0$ is joined to x_0 by a whole sphere of minimising geodesics, so $B(\bar{x}_0, \pi) = \mathbb{S}^n \setminus \{x_0\}$ is a free ball and $\mathcal{R}_X^g = \mathbb{S}^n \setminus \{x_0\}$ while $\mathcal{R}_X = \emptyset$. The compact reconstruction theorem below must therefore be stated for M_X^g , not M_X .

2.2 The direction field

Lemma 2.1 (Direction field: compactness and semicontinuity). *For every $x \in M \setminus X$ the set $U(x)$ is nonempty and compact, and the multimap $x \mapsto U(x)$ is upper semicontinuous: if $x_k \rightarrow x$ in $M \setminus X$ and $u_k \in U(x_k)$, $u_k \rightarrow u$ (in TM), then $u \in U(x)$. In particular U is single-valued and continuous at every point of $(M \setminus X) \setminus \Sigma$.*

Proof. Nonemptiness and compactness: minimisers exist by properness, and a limit of initial vectors of minimising geodesics from the same x is the initial vector of a limit geodesic, which minimises by continuity of d_X and of lengths. Upper semicontinuity: let γ_k be the minimising geodesics with $\gamma'_k(0) = u_k$. By smooth dependence of geodesics on initial conditions, $\gamma_k \rightarrow \gamma$ locally uniformly where $\gamma'(0) = u$; the lengths $d_X(x_k) \rightarrow d_X(x)$, and the endpoints $\gamma_k(d_X(x_k)) \in X$ converge to $\gamma(d_X(x)) \in X$ (X closed). So γ is a geodesic from x to X of length $d_X(x)$: minimising, i.e. $u \in U(x)$. At a point with $U(x) = \{u\}$, upper semicontinuity of a compact-valued multimap into a singleton forces continuity of every selection near x — note this is continuity *at* x only; U may be multivalued arbitrarily close by. $\square$

Lemma 2.2 (First variation of d_X ; [8, §3]). *For $x \in M \setminus X$ and $w \in T_x M$, the one-sided directional derivative of d_X exists and equals*

$$\left. \frac{d}{ds} \right|_{s=0^+} d_X(\exp_x(sw)) = \min_{u \in U(x)} \langle w, -u \rangle.$$

Proof. “ $\leq$ ”: fix $u \in U(x)$ with geodesic γ , and fix a small $\delta > 0$. The point $\gamma(\delta)$ lies in a normal ball around x , so $q \mapsto d(q, \gamma(\delta))$ is smooth near x with gradient $-u$ at x . Since points of γ satisfy $d_X(\gamma(\delta)) = d_X(x) - \delta$,

$$d_X(\exp_x(sw)) \leq d(\exp_x(sw), \gamma(\delta)) + d_X(x) - \delta = d_X(x) + s \langle w, -u \rangle + O_\delta(s^2),$$

so $\limsup_{s \downarrow 0} s^{-1}[d_X(\exp_x(sw)) - d_X(x)] \leq \langle w, -u \rangle$ for every $u \in U(x)$, hence $\leq$ the minimum.

“ $\geq$ ”: let $s_k \downarrow 0$ realise the $\liminf$, put $x_k = \exp_x(s_k w)$, pick $u_k \in U(x_k)$ with geodesics γ_k , and set $\delta_k = \sqrt{s_k}$. The broken path through $\gamma_k(\delta_k)$ gives $d_X(x) \leq d(x, \gamma_k(\delta_k)) + d_X(x_k) - \delta_k$, and in normal coordinates at x_k , where $v_k := \exp_{x_k}^{-1}(x) = -s_k w + O(s_k^2)$,

$$d(x, \gamma_k(\delta_k)) \leq |\delta_k u_k - v_k| + O(\delta_k^3) \leq \delta_k - \langle v_k, u_k \rangle + \frac{|v_k|^2}{2\delta_k} + O(\delta_k^3) = \delta_k + s_k \langle w, u_k \rangle + O(s_k^{3/2}),$$

using $\sqrt{1+\tau} \leq 1 + \tau/2$ and the $O(r^2)$ metric distortion of normal coordinates. Hence $d_X(x_k) - d_X(x) \geq s_k \langle w, -u_k \rangle - O(s_k^{3/2})$. By Lemma 2.1 a subsequence of u_k converges to some $u \in U(x)$, so the $\liminf$ is $\geq \langle w, -u \rangle \geq \min_{u \in U(x)} \langle w, -u \rangle$. (Cf. [8, §3], where the semiconcavity of d_X on $M \setminus X$ — not needed here — is also established.) $\square$

Intuitively: d_X increases fastest, at unit rate, in the direction directly *away* from a nearest foot; at a two-geodesic point the rate in a given direction w is dictated by the *least favourable* competing foot, which is what makes the $\min$ appear and what makes the gradient flow turn.

2.3 The ascent inclusion and capture curves

On the open set $M \setminus X$ define the multimap

$$G(x) := \text{conv}(-U(x)) \subseteq T_x M,$$

the convex hull of the reversed minimising directions. By Lemma 2.1, G is upper semicontinuous with nonempty compact convex values contained in the closed unit ball. A *capture curve* from $p \in M \setminus X$ is a locally Lipschitz curve $\Phi : [0, T_{\max}) \rightarrow M \setminus X$ with $\Phi_0 = p$ and

$$\Phi'(t) \in G(\Phi_t) \quad \text{for a.e. } t, \tag{1}$$

maximally extended in $M \setminus X$. Local existence of solutions of (1), and extension as long as the trajectory stays in a compact subset of the open domain, are the standard theory of differential inclusions with upper semicontinuous convex compact right-hand side (Aubin–Cellina [3, §2.1]), applied in charts and concatenated; $|\Phi'| \leq 1$ a.e., so trajectories are 1-Lipschitz. We emphasise that *no* uniqueness, canonicity, or semiconcave-gradient selection is needed: any solution serves. (At a point of Σ the inclusion may branch; off Σ it is the classical ODE of the field $-u$.) When U is single-valued along a trajectory the curve simply extends the current minimising geodesic beyond x , switching fields continuously as the foot slides.

2.4 Free-ball functions and medial limit functions

For $a \in M \setminus X$ set

$$\psi_a := d(\cdot, a) - d_X(a) : M \rightarrow \mathbb{R},$$

a 1-Lipschitz function, negative exactly on the open free ball $B(a, d_X(a))$, with $\psi_a \geq 0$ on X . It is normalised by the reach — $\min \psi_a = -d_X(a)$, attained at a — rather than by a basepoint; this intrinsic normalisation, “measure the ball by how deep it goes, not by how far its centre is”, is what makes the limits below basepoint-free.

Definition 2.3 (Medial limit functions). *A function $\psi : M \rightarrow \mathbb{R}$ is a medial limit function of X if there are $a_k \in M_X^g$ with $\psi_{a_k} \rightarrow \psi$ locally uniformly. It is finite if (a_k) has a bounded subsequence and ideal otherwise (then ψ is a horofunction-type limit: $d(a_k, \cdot) \rightarrow \infty$ locally uniformly while ψ stays finite). Every ψ is 1-Lipschitz, and $\psi \geq 0$ on X (each ψ_{a_k} is), so $\{\psi < 0\}$ is an open set disjoint from X : the free horoball of ψ . For finite ψ along $a_k \rightarrow a$: if $a \in M_X^g$ then $\psi = \psi_a$ is a free-ball function; in general (M_X^g is not closed) $\psi = \psi_a$ with a a focal-degenerate limit of medial centres, or even a point of X itself (medial centres collapsing into a corner of X with $d_X(a_k) \rightarrow 0$), where $\psi_a = d(\cdot, a)$ and the free horoball is empty — still usable, by Lemma 4.1, which never needs the limit centre itself.*

3 The capture theorem

The capture theorem is the engine of the compact case and one of the two inputs to the general characterisation. Its statement is a clean dichotomy; its proof is a one-line differential identity valid off the axis, plus a compactness argument for the escape alternative.

Theorem 3.1 (Capture or escape). *Let M be complete, X closed, $p \in M \setminus X$, and let Φ be any capture curve from p ; assume $p \notin \Sigma$ (otherwise p is captured at time 0: $a = p$ contains p at depth $d_X(p)$ in its own medial ball). Let*

$$T := \sup\{t \in [0, T_{\max}) : \Phi_s \notin \Sigma \ \forall s \in [0, t]\} \in [0, T_{\max}]$$

($T = 0$ is possible: Σ need not be closed, and the trajectory may approach it instantly; this lands in case (i)). Then on $[0, T)$,

$$d_X(\Phi_t) = d_X(p) + t, \quad d(p, \Phi_t) \leq t, \quad \text{hence} \quad d_X(\Phi_t) - d(p, \Phi_t) \geq d_X(p), \quad (2)$$

and exactly one of the following holds.

- (i) (Capture.) $T < T_{\max}$. Then there exist $a \in M_X^g$ arbitrarily close to Φ_T ($a = \Phi_T$ itself if $\Phi_T \in \Sigma$) with

$$d_X(a) - d(p, a) \geq d_X(p) - \varepsilon \quad (\varepsilon > 0 \text{ arbitrary}),$$

so $p \in B(a, d_X(a))$ for such a : $p \in \mathcal{R}_X^g$, at depth arbitrarily close to $d_X(p)$, and $T \leq \text{diam}(M) - d_X(p)$.

- (ii) (Escape.) $T = T_{\max}$. Then $T_{\max} = \infty$, the trajectory never meets Σ , $d_X(\Phi_t) = d_X(p) + t \rightarrow \infty$, and Φ_t leaves every compact subset of M . This is possible only if M is non-compact.

Proof. The rate-one identity. Fix $t \in [0, T]$; on $[0, t]$ the trajectory avoids Σ , so $U(\Phi_s) = \{u(\Phi_s)\}$ is a singleton for every $s \leq t$ and (1) reads $\Phi'(s) = -u(\Phi_s)$ at a.e. s . The map $h(s) := d_X(\Phi_s)$ is Lipschitz. Let s be any point where both $\Phi'(s)$ exists (with the inclusion valid) and $h'(s)$ exists — almost every $s \in [0, t]$. Write $v = \Phi'(s) = -u(\Phi_s)$, a unit vector. In a chart around Φ_s , differentiability gives $d(\Phi_{s+\sigma}, \exp_{\Phi_s}(\sigma v)) = o(\sigma)$ as $\sigma \downarrow 0$, and d_X is 1-Lipschitz, so

$$h(s + \sigma) = d_X(\exp_{\Phi_s}(\sigma v)) + o(\sigma) = h(s) + \sigma \min_{u \in U(\Phi_s)} \langle v, -u \rangle + o(\sigma) = h(s) + \sigma \langle -u, -u \rangle + o(\sigma),$$

by Lemma 2.2 and single-valuedness. Hence $h'(s) = |u|^2 = 1$ at a.e. $s \in [0, t]$, and integrating, $h(t) = d_X(p) + t$. The bound $d(p, \Phi_t) \leq t$ is the 1-Lipschitz property of the trajectory, and the margin inequality in (2) is their difference. Note that no closedness of Σ , and no structure at focal or conjugate points, was used: a point of $(M \setminus X) \setminus \Sigma$ in the closure of Σ is simply a point where U is single-valued, and the identity passes through it.

Dichotomy. Suppose first $T < T_{\max}$. If $\Phi_T \in \Sigma$, continuity of $d_X(\Phi) - d(p, \Phi)$ and (2) on $[0, T]$ give $d_X(\Phi_T) - d(p, \Phi_T) \geq d_X(p) > 0$, so $a = \Phi_T \in M_X^g$ contains p in its open medial ball at depth $\geq d_X(p)$. If $\Phi_T \notin \Sigma$, then by definition of T as a supremum, every interval $(T, T + \delta] \subseteq [0, T_{\max})$ contains a time s_δ with $\Phi_{s_\delta} \in \Sigma$; letting $\delta \downarrow 0$ and using continuity of the margin, $a = \Phi_{s_\delta} \in M_X^g$ satisfies $d_X(a) - d(p, a) \geq d_X(p) - \varepsilon$ for δ small. In both cases $p \in B(a, d_X(a))$ with $a \in M_X^g$, i.e. $p \in \mathcal{R}_X^g$; and $d_X \leq \text{diam}(M)$ bounds T via (2).

Now suppose $T = T_{\max}$: the trajectory never meets Σ , and (2) holds on all of $[0, T_{\max})$. If $T_{\max} < \infty$, then $\Phi([0, T_{\max})) \subseteq \bar{B}(p, T_{\max}) \cap \{d_X \geq d_X(p)\}$, a compact subset of the open set $M \setminus X$ (properness of M); by the extension property of solutions this contradicts maximality. So $T_{\max} = \infty$ and $d_X(\Phi_t) = d_X(p) + t \rightarrow \infty$; since d_X is bounded on compact sets, Φ_t leaves every compact set. On compact M , $d_X \leq \text{diam}(M)$ makes this impossible, so case (i) occurs. $\square$

Corollary 3.2 (Total reconstruction on compact manifolds, unconditional). *Let M be a compact Riemannian manifold and $X \subseteq M$ closed, $\emptyset \neq X \neq M$. Then*

$$\boxed{\mathcal{R}_X^g = M \setminus X},$$

and quantitatively: every $p \in M \setminus X$ lies at depth $\geq d_X(p) - \varepsilon$ (any $\varepsilon > 0$) inside a medial ball of M_X^g . No curvature, genericity, analyticity, spread, or cardinality hypothesis is required; in particular $|X| = 1$ is allowed, where the two-point axis has $M_X = \emptyset$ but the two-geodesic axis still reconstructs everything.

Proof. $\mathcal{R}_X^g \subseteq M \setminus X$ since medial balls are free. Conversely, if $p \in M_X^g$ then $p \in B(p, d_X(p)) \subseteq \mathcal{R}_X^g$; otherwise start a capture curve at p (local existence, §2); by Theorem 3.1 escape is impossible on compact M , so p is captured. $\square$

In particular the reconstruction question on compact Riemannian manifolds, raised for the medial axis in [4, §5.2], has an unconditional affirmative answer for the two-geodesic axis.

Corollary 3.3 (Two-point axis under spread). *Let M be complete with $\text{inj}(M) := \inf_{q \in M} \text{inj}(q) > 0$, and suppose*

$$\sup_{M \setminus X} d_X < \text{inj}(M) \quad \text{— e.g. } X \text{ } \delta\text{-dense with } \delta < \text{inj}(M).$$

Then $\mathcal{R}_X = M \setminus X$ for the two-point axis. Compactness is not needed: it would enter only to preclude escape, which the spread hypothesis already does. In $\mathbb{R}^n$ ($\text{inj} = \infty$) the statement recovers the classical fact that every coarsely dense closed set reconstructs its complement: $\sup d_X < \infty$ implies $\mathcal{R}_X = \mathbb{R}^n \setminus X$.

Proof. Escape is impossible: along an escaping capture curve, $d_X(\Phi_t) = d_X(p) + t \rightarrow \infty$ would exceed $\sup d_X$. So every $p \in (M \setminus X) \setminus M_X^g$ is captured (Theorem 3.1(i), within time $\sup d_X - d_X(p)$), giving $a \in M_X^g$ with $p \in B(a, d_X(a))$; for $p \in M_X^g$ take $a = p$. Two distinct minimising geodesics from a with the same endpoint x_0 would give two distinct vectors of norm $d(a, x_0) = d_X(a) < \text{inj}(M) \leq \text{inj}(a)$ with the same image under $\exp_a$, contradicting injectivity of $\exp_a$ on the open ball of radius $\text{inj}(a)$. So the two geodesics of a have distinct endpoints: $a \in M_X$. $\square$

The formulation *stops* at the axis rather than flowing through it, and this is essential. The margin $d_X(\Phi) - d(p, \Phi)$ one would like to accumulate is *not* monotone through multi-geodesic arcs — already for two points in $\mathbb{R}^2$, where the flow reaches the bisector and slides along it while the margin strictly decreases toward the depth of p in the limiting half-plane — so no useful quantity survives a passage *through* Σ . But Σ is the goal, not an obstacle: stopping at the first approach both sidesteps the non-monotonicity and delivers capture with the margin (2) intact, and it makes the argument indifferent to the failure of Σ to be closed — focal degenerations are unique-geodesic points, through which the rate-one identity passes unharmed. This is what frees the theory from every regularity question at cut, focal, and conjugate points.

4 Drift: the swallow, the characterisation, the trichotomy

Capture is complete on compact manifolds, but on a non-compact M a point can be reconstructed by medial balls whose centres go to infinity while its own capture curve escapes. The primal example is $X = \{x_0, x_1\} \subseteq \mathbb{R}^2$: a point p in an open half-plane bounded by the line x_0x_1 , placed beyond x_0 at an obtuse angle, has a capture curve that runs straight to infinity with the single foot x_0 — yet p is swallowed by remote bisector balls $B(a_k, r(a_k))$: a single sufficiently distant ball already contains it, and so does every later member of the escaping sequence $a_k \rightarrow \infty$. The limit of those balls is a half-plane, and the correct general home for such limits is the space of medial limit functions of Definition 2.3.

Lemma 4.1 (Swallow, general and offset-free). *Let M be any complete Riemannian manifold, X closed, and ψ a medial limit function of X . Then*

$$\{\psi < 0\} \subseteq \mathcal{R}_X^g.$$

If the witnessing centres lie in M_X , then $\{\psi < 0\} \subseteq \mathcal{R}_X$.

Proof. Let $\psi = \lim \psi_{a_k}$ locally uniformly, $a_k \in M_X^g$, and let $q \in \{\psi < 0\}$. Then $\psi_{a_k}(q) \rightarrow \psi(q) < 0$, so for large k

$$d(q, a_k) - d_X(a_k) = \psi_{a_k}(q) < 0,$$

i.e. $q \in B(a_k, d_X(a_k))$ with $a_k \in M_X^g$: $q \in \mathcal{R}_X^g$. $\square$

There is no anchoring of feet, no curvature bound, no pinching, no offset hypothesis: the entire content of the swallowing step — which in curved settings historically required delicate control of where the escaping centres' feet accumulate — is absorbed into the normalisation of ψ_a by the reach.

Lemma 4.2 (Compactness of reconstructing sequences). *Let $p \in \mathcal{R}_X^g$, and let $a_k \in M_X^g$ be any sequence with $\limsup_k [d_X(a_k) - d(p, a_k)] > 0$ (e.g. a sequence of centres whose balls contain p with margin bounded away from 0; a single ball repeated qualifies). Then a subsequence of (ψ_{a_k}) converges locally uniformly to a medial limit function ψ with $\psi(p) < 0$.*

Proof. The functions ψ_{a_k} are 1-Lipschitz and $\psi_{a_k}(p) = d(p, a_k) - d_X(a_k)$ is bounded: below by $-d_X(p)$ (since $d_X(a_k) \leq d_X(p) + d(p, a_k)$) and above by 0 along the qualifying subsequence. By Arzelà–Ascoli on the proper space M , a further subsequence converges locally uniformly; the limit is a medial limit function, and $\psi(p) = \lim \psi_{a_k}(p) \leq -\limsup [d_X(a_k) - d(p, a_k)] < 0$. $\square$

Theorem 4.3 (Characterisation on an arbitrary complete manifold). *For every complete Riemannian manifold M and every closed $X \subseteq M$, $\emptyset \neq X \neq M$,*

$$\mathcal{R}_X^g = \bigcup \{ \{\psi < 0\} : \psi \text{ a medial limit function of } X \}.$$

The same identity holds for $\mathcal{R}_X$ and M_X in place of $\mathcal{R}_X^g$ and M_X^g . (Every finite medial limit function is itself of the form $\psi_a = d(\cdot, a) - d_X(a)$ with a in the closure of M_X^g taken in M : a focal-degenerate limit of medial centres, or — degenerately — a point of X , where $d_X(a) = 0$ and the sublevel set is empty. Either way its sublevel set is swallowed by the lemma, which never needs the limit centre itself.)

Proof. “ $\supseteq$ ” is Lemma 4.1. “ $\subseteq$ ”: if $p \in \mathcal{R}_X^g$ then $p \in B(a, d_X(a))$ for some $a \in M_X^g$, and ψ_a — a medial limit function via the constant sequence — has $\psi_a(p) < 0$. $\square$

The identity is deliberately economical, and one should read its force correctly. Its finite members alone give a tautology — every ball is its own limit — so the set equality is not the point. The point is the pair of facts it packages. *First*, the *ideal* members reconstruct their entire open sublevel set, bare, with no anchoring or pinching hypothesis: this is Lemma 4.1 applied to horofunction-type limits. *Second*, combined with Theorem 3.1 it yields a clean trichotomy.

Theorem 4.4 (Trichotomy on a complete manifold). *Let M be complete, X closed, $p \in M \setminus X$. Exactly one of the following holds.*

- (1) (Captured.) *Some capture curve from p approaches M_X^g in finite time. Then $p \in \mathcal{R}_X^g$, at depth arbitrarily close to $d_X(p)$.*
- (2) (Swallowed, uncaptured.) *No capture curve from p approaches M_X^g in finite time, but $\psi(p) < 0$ for some medial limit function ψ . Then $p \in \mathcal{R}_X^g$: reconstructed, but only by centres p 's own ascent cannot find — a remote ball, or a drifting sequence.*

- (3) (Escaping.) *Neither. Then $p \notin \mathcal{R}_X^g$, and every capture curve from p escapes to infinity with $d_X(\Phi_t) = d_X(p) + t$.*

In particular $\mathcal{R}_X^g = \{p : (1) \text{ or } (2)\}$, and on compact M class (1) is all of $M \setminus X$.

Proof. The classes are exclusive and exhaustive by construction. (1) $\Rightarrow p \in \mathcal{R}_X^g$ is Theorem 3.1(i); (2) $\Rightarrow p \in \mathcal{R}_X^g$ is Lemma 4.1. Conversely, if $p \in \mathcal{R}_X^g$ then $p \in B(a, d_X(a))$ for some $a \in M_X^g$ and the finite medial limit function ψ_a has $\psi_a(p) < 0$, so p lies in class (2) whenever it is not in class (1) — never in class (3). Finally, a point not in class (1) has every capture curve uncaptured, hence escaping, by the dichotomy of Theorem 3.1. $\square$

Białożył's theorem is not a description of the reconstructed set but a *reconstructibility criterion*: it says exactly when a closed set is recovered in full. The trichotomy yields the direct analogue on any complete manifold, because $\mathcal{R}_X^g$ is precisely classes (1)–(2) while class (3) is its complement.

Corollary 4.5 (Reconstructibility criterion). *Let M be complete, X closed, $\emptyset \neq X \neq M$. The following are equivalent.*

- (a) *X is reconstructible: $\mathcal{R}_X^g = M \setminus X$.*
- (b) *$M \setminus X \subseteq \bigcup \{\psi < 0\} : \psi \text{ a medial limit function of } X\}$.*
- (c) *Every $p \in M \setminus X$ is captured or drift-swallowed; equivalently, the escaping class (3) of Theorem 4.4 is empty.*
- (d) *For every $p \in M \setminus X$ whose capture curves escape to infinity, some medial limit function ψ satisfies $\psi(p) < 0$.*

Proof. (a) $\Leftrightarrow$ (b): by Theorem 4.3 the right-hand side of (b) equals $\mathcal{R}_X^g$, and $\mathcal{R}_X^g \subseteq M \setminus X$ always, so (b) is the reverse inclusion, i.e. (a). (a) $\Leftrightarrow$ (c): by Theorem 4.4, $\mathcal{R}_X^g$ is exactly classes (1)–(2) and is disjoint from class (3); since the three classes partition $M \setminus X$, $\mathcal{R}_X^g = M \setminus X$ iff class (3) is empty. (c) $\Leftrightarrow$ (d): a point outside class (1) escapes (Theorem 3.1), so it lies in class (2) — swallowed — iff some ψ has $\psi(p) < 0$; captured points lie in $\mathcal{R}_X^g$ regardless, so emptiness of class (3) is exactly (d). $\square$

Two points fix the meaning of the criterion. *On a compact manifold it is vacuous*: escape is impossible (Theorem 3.1), class (3) is empty, and X is reconstructible for free (Corollary 3.2); it therefore has content only on non-compact M , exactly as Białożył's does. And condition (b), which quantifies over the medial limit functions and hence implicitly over the axis M_X^g , is the *unconditional* form; its *faithful* sharpening — a certificate readable from X alone, with the axis absent — is Białożył's theorem in the flat case, which we recover now.

Corollary 4.6 (Flat specialisation: Białożył's theorem). *In $\mathbb{R}^n$ geodesics between points are unique, so $M_X^g = M_X$ and $\mathcal{R}_X^g = \mathcal{R}_X$. A finite medial limit function is a free-ball function $y \mapsto |y - a| - r$; an ideal one, formed along centres $a_k \rightarrow \infty$ in a unit direction u with reaches r_k , is affine,*

$$\psi(y) = \lim_k (|y - a_k| - r_k) = \beta - \langle y, u \rangle, \quad \{\psi < 0\} = \{\langle y, u \rangle > \beta\},$$

an open half-space with outward normal u (the offset $|a_k| - r_k \rightarrow \beta$ by convergence of ψ_{a_k}). Since the finite balls fill $\overline{\text{conv}} X \setminus X$ and the ideal half-space limits are exactly the outer half-spaces L^+

of the defective supporting hyperplanes — those with $X \cap L \neq \overline{\text{conv}} X \cap L$, equivalently (Motzkin [10]) carrying a non-Chebyshev point just outside the contact face [4] — Theorem 4.3 specialises to

$$\mathcal{R}_X = (\overline{\text{conv}} X \setminus X) \cup \bigcup_{L \text{ defective}} L^+,$$

and the reconstructibility criterion it induces is exactly Theorem 1.1; the escaping class (3) of Theorem 4.4 is Białożyty’s unreconstructed set. Thus the abstract characterisation supplies “ $\mathcal{R}_X$ is balls $\cup$ half-space limits”, and Białożyty’s certificate identifies the half-space limits as the defective supporting hyperplanes.

5 The homotopy layer: capture, the Aubry set, and the limits of swallowing

The capture theorem is silent about homotopy type. Its argument runs the ascent inclusion only off Σ , imposes no canonical selection, and *terminates* at the axis — the opposite of the singular gradient flow of Lieutier [7] and Albano–Cannarsa–Nguyen–Sinestrari [2], whose purpose is to flow *into* the singular set and deduce homotopy equivalence. This section imports that flow and settles the homotopy question for the completed theory: when the axis carries the homotopy type of the complement, what obstructs it when it does not, and what either answer has to do with the reconstruction $\mathcal{R}_X^g$. Throughout, Φ denotes the *canonical* flow: the unique generalised-gradient trajectory of d_X through each point of $M \setminus X$ — Lieutier’s flow in $\mathbb{R}^n$ [7]; existence and uniqueness [2, Thm. 3.4, Cor. 3.6] in the Euclidean case and [2, §4] on Riemannian domains. The construction is local, its only input being the local semiconcavity of d_X on $M \setminus X$ [1, 8], so it defines Φ for every closed X in every complete M ; uniqueness gives the semigroup property. Canonical trajectories satisfy the inclusion (1), so Theorem 3.1 applies to them verbatim; and they are global in time: d_X is nondecreasing along them, so they stay in $M \setminus X$, and speed ≤ 1 together with completeness of M rules out finite-time breakdown. We import two further facts, both local, from that literature.

- (F1) *Joint continuity*: $(t, p) \mapsto \Phi_t(p)$ is continuous on $[0, \infty) \times (M \setminus X)$. (Local semiconcavity makes the ascending field one-sidedly Lipschitz, and Grönwall gives locally Lipschitz dependence on the initial point; this is the continuity that underlies the homotopy statements of [7, 2].)
- (F2) *Propagation*: Σ is positively invariant under Φ : once a canonical trajectory is singular it stays singular [2, Thms. 3.7 and 4.5]. (Stated there for domains, under curvature bounds; the proof — semiconcavity plus a differential inequality — is local, and a finite time window of a trajectory lives in a compact subset of $M \setminus X$, where the bounds hold. So it applies verbatim here.)

Lemma 5.1 (Capture is permanent; the cut locus is invariant). *Let $p \in M \setminus X$ and suppose the canonical trajectory of p is captured, with first-approach time T as in Theorem 3.1(i). Then*

$$\Phi_t(p) \in \Sigma \quad \text{for every } t > T :$$

the trajectory does not merely approach the axis; it enters it and never leaves. Moreover the cut locus

$$\text{Cut}(X) := \overline{\Sigma} \cap (M \setminus X) \quad (\text{closure in } M)$$

is positively invariant as well.

Proof. If $\Phi_T \in \Sigma$, apply (F2) from time T . If $\Phi_T \notin \Sigma$, then by the definition of T every interval $(T, T+\delta]$ contains a time s_δ with $\Phi_{s_\delta} \in \Sigma$ (proof of Theorem 3.1); (F2) from time s_δ gives $\Phi_t \in \Sigma$ for all $t \geq s_\delta$, and $\delta \downarrow 0$ covers all of (T, ∞) . For the cut locus: $q \in \text{Cut}(X)$ is a limit of $a_j \in \Sigma$, so by (F1) and (F2), $\Phi_t(q) = \lim_j \Phi_t(a_j) \in \bar{\Sigma}$; and $\Phi_t(q) \in M \setminus X$ always. $\square$

Theorem 5.2 (Uniform capture is a homotopy equivalence). *Let M be complete, X closed, and let $U \subseteq M \setminus X$ be open and positively invariant under the canonical flow, on which capture is uniform: for some $D < \infty$ the trajectory of every $p \in U$ meets Σ by time D . Then $\Phi_D(U) \subseteq \Sigma \cap U$, and the inclusions*

$$\Sigma \cap U \hookrightarrow \text{Cut}(X) \cap U \hookrightarrow U$$

are homotopy equivalences, with Φ_D as homotopy inverse and the flow itself as the homotopy.

Proof. $\Phi_D(p) \in \Sigma$ for $p \in U$ by Lemma 5.1 (the trajectory meets Σ at some $t^* \leq D$ and stays), and $\Phi_D(p) \in U$ by invariance. Set $r := \Phi_D: U \rightarrow \Sigma \cap U$, write $i: \Sigma \cap U \hookrightarrow U$ for the inclusion, and let $H(p, s) := \Phi_{sD}(p)$, jointly continuous by (F1). Then H is a homotopy in U from id_U to $i \circ r$; and H restricted to $(\Sigma \cap U) \times [0, 1]$ takes values in $\Sigma \cap U$ ((F2) and invariance of U), so it is also a homotopy from $\text{id}_{\Sigma \cap U}$ to $r \circ i$. Hence i is a homotopy equivalence. The same two sentences with $\text{Cut}(X) \cap U$ in place of $\Sigma \cap U$ — using the invariance of the cut locus from Lemma 5.1 and $\Phi_D(U) \subseteq \Sigma \cap U \subseteq \text{Cut}(X) \cap U$ — show that $\text{Cut}(X) \cap U \hookrightarrow U$ is a homotopy equivalence, and the inner inclusion follows by two-out-of-three. $\square$

Corollary 5.3 (Bounded distance: topological faithfulness). *Let M be complete, X closed, $\emptyset \neq X \neq M$, and suppose $D := \sup_{M \setminus X} d_X < \infty$. Then escape is impossible, $\mathcal{R}_X^g = M \setminus X$, every component Ω of $M \setminus X$ meets Σ , and*

$$\Sigma \cap \Omega \hookrightarrow \text{Cut}(X) \cap \Omega \hookrightarrow \Omega$$

are homotopy equivalences. In particular:

- (a) *on every compact M : $M \setminus X \simeq M_X^g \simeq \text{Cut}(X)$, unconditionally — the closed-set counterpart of the classical retraction of $M \setminus \{x_0\}$ onto $\text{Cut}(x_0)$;*
- (b) *for every coarsely dense closed set in any complete M — e.g. a periodic $X \subseteq \mathbb{R}^n$ — the complement, an unbounded domain outside the scope of [7, 2], has the homotopy type of its two-geodesic axis and of its cut locus.*

With Corollary 3.3 this completes a pairing: $\sup d_X < \text{inj}(M)$ makes the two-point reconstruction metrically complete ($\mathcal{R}_X = M \setminus X$), and $\sup d_X < \infty$ alone makes the two-geodesic picture topologically faithful.

Proof. Escape forces $d_X(\Phi_t) = d_X(p) + t \rightarrow \infty$ (Theorem 3.1(ii)), impossible under the bound; so every $p \notin M_X^g$ is captured, hence $p \in \mathcal{R}_X^g$ (Theorem 3.1(i)), and $M_X^g \subseteq \mathcal{R}_X^g$ trivially: $\mathcal{R}_X^g = M \setminus X$. If a canonical trajectory avoided Σ on $[0, \tau]$, the rate-one identity (2) would give $d_X(p) + \tau \leq D$; hence every trajectory meets Σ by time $D - d_X(p) \leq D$, and the meeting point lies in the component of its start. Apply Theorem 5.2 with $U = \Omega$ (open; invariant, since trajectories are continuous and stay in $M \setminus X$). Compact M has $D \leq \text{diam}(M)$; a δ -dense X has $D \leq \delta$. $\square$

Beyond the bound, escape occurs and the statement genuinely changes. The right language is the *Aubry set* of weak KAM theory, and the translation is exact.

Lemma 5.4 (The Aubry set of d_X is the escape class). *Following [5, Def. 1.4], let*

$$\mathcal{A}^*(X) := \left\{ q \in M \setminus X : q = \gamma(t_0) \text{ for some } t_0 > 0 \right. \\ \left. \text{and some arclength } \gamma : [0, \infty) \rightarrow M \text{ with } d_X(\gamma(t)) = t \ \forall t \geq 0 \right\},$$

the union of the infinite calibrated rays of X . Then:

- (i) $\mathcal{A}^*(X) \cap \Sigma = \emptyset$;
- (ii) $\mathcal{A}^*(X)$ is closed in $M \setminus X$;
- (iii) a point $p \in M \setminus X$ lies in $\mathcal{A}^*(X)$ if and only if its capture curves escape — that is, $\mathcal{A}^*(X)$ is exactly classes (2) $\cup$ (3) of Theorem 4.4, and

$$M \setminus (X \cup \mathcal{A}^*(X)) = \{\text{captured points}\} \supseteq \Sigma,$$

an open set.

Proof. (i) Let $q = \gamma(t_0)$ lie on a calibrated ray. If q had a second minimising geodesic $\sigma \neq \gamma|_{[0, t_0]}$, the curve σ followed by $\gamma|_{[t_0, t]}$ would realise $d_X(\gamma(t)) = t$ for $t > t_0$ with a corner at q , hence could be strictly shortened — contradiction. (The same argument shows that past a point of Σ no minimising geodesic keeps minimising, so calibrated rays avoid Σ outright.)

(ii) Let $q_j = \gamma_j(t_j) \rightarrow q \in M \setminus X$ with γ_j calibrated. Then $t_j = d_X(q_j) \rightarrow d_X(q) > 0$, and the feet $\gamma_j(0)$ eventually lie in the compact set $\bar{B}(q, 2d_X(q)) \cap X$. Calibrated rays are unit-speed minimising geodesics on every initial segment ([5, Rem. 1.5]); by properness of M a subsequence converges locally uniformly to an arclength ray γ , and $d_X(\gamma(t)) = t$ passes to the limit: $q = \gamma(d_X(q)) \in \mathcal{A}^*(X)$.

(iii) All capture curves from $p \notin \Sigma$ agree until their first approach to Σ : off Σ the inclusion (1) is the ODE of the single unit field, and the trajectory is the arclength extension of the minimising geodesic of p (§2). If that trajectory escapes, it never meets Σ and the rate-one identity holds forever; prepending the minimising segment from the foot of p produces a curve γ with $d_X(\gamma(t)) = t$ for all $t \geq 0$: an infinite calibrated ray through p , so $p \in \mathcal{A}^*(X)$. Conversely let $p = \gamma(t_0) \in \mathcal{A}^*(X)$; by (i) applied at each $\gamma(s)$, $s > 0$, the ray avoids Σ , and since the trajectory off Σ is the extension of the unique minimising geodesic, the trajectory of p is $t \mapsto \gamma(t_0 + t)$: it never meets Σ , and $d_X \rightarrow \infty$ along it — escape. Points of Σ are captured at time 0 and excluded from $\mathcal{A}^*(X)$ by (i); openness of the captured set is (ii). $\square$

Theorem 5.5 (Cannarsa–Cheng–Fathi [5, Thm. 1.6], in the vocabulary of the trichotomy). *For every closed $X \neq \emptyset$ in every complete M , the inclusion*

$$M_X^g \hookrightarrow M \setminus (X \cup \mathcal{A}^*(X)) = \{\text{captured points}\}$$

is a homotopy equivalence: the captured part of the complement always has the homotopy type of the axis — no bound on d_X , no compactness. (Under uniform capture this is Theorem 5.2 with $U = M \setminus X$; in general the uniform clock fails and the statement is exactly [5, Thm. 1.6], translated by Lemma 5.4. Their Theorem 1.1 adds that M_X^g is always locally contractible.)

6 The medial complex: the ideal cells close the leak

Homotopy type is transported by capture and by nothing else (Theorem 5.5): the axis sees the captured class (1) but not the Aubry set (2) $\cup$ (3). Swallowing, metrically complete as it may be, transports nothing — a closed set can be reconstructed in full while its complement and its axis have different homotopy types, already for three points in the plane, whose complement is a wedge of three circles facing a contractible Voronoi tripod. This section repairs the leak, on every complete manifold, for every closed X , with no hypothesis. The entire content is a covering theorem: *the finite layer does not leak* — every swallowed-uncaptured point lies in an *ideal* free horoball, at depth no smaller than its depth in any finite medial ball (Theorem 6.3). Granting it, the captured set and the ideal free horoballs form an open cover of $\mathcal{R}_X^g$ whose homotopy colimit is $\mathcal{R}_X^g$ by the gluing calculus of Dugger–Isaksen [6], and the captured vertex is identified with M_X^g by Theorem 5.5.

Throughout, $W := M \setminus (X \cup \mathcal{A}^*(X))$ denotes the captured set, open by Lemma 5.4, and for $p \in M \setminus X$ and $a \in M$ we write

$$u_p(a) := d_X(a) - d(a, p),$$

the *margin of p at a* : $u_p(a) > 0$ with $a \in M_X^g$ says exactly that the medial ball of a swallows p at depth $u_p(a)$. As a difference of 1-Lipschitz functions, u_p is 2-Lipschitz in a (and in p).

Lemma 6.1 (Margin transport). *Let $x \in M \setminus X$, let Φ be a capture curve from x , and let $T = \sup\{t : \Phi_s \notin \Sigma \ \forall s \leq t\}$ be its first approach to the axis as in Theorem 3.1. Then for every $p \in M$ the margin $t \mapsto u_p(\Phi_t)$ is nondecreasing on $[0, T]$ and 2-Lipschitz on all of $[0, T_{\max}]$:*

$$u_p(\Phi_t) \geq u_p(x) \quad (t \leq T), \quad |u_p(\Phi_t) - u_p(\Phi_s)| \leq 2|t - s|.$$

Moreover, on any interval $[0, t]$ on which Φ avoids Σ , the curve is the arclength extension of the minimising geodesic of x : capture curves are unique, and are calibrated geodesics, until their first approach.

Proof. On $[0, T]$ the rate-one identity (2) gives $d_X(\Phi_t) = d_X(x) + t$, while $d(\Phi_t, p) \leq d(x, p) + t$ since Φ is 1-Lipschitz; subtracting, $u_p(\Phi_t) \geq u_p(x)$, and continuity extends this to $t = T$. The Lipschitz bound is the 2-Lipschitz property of u_p along a 1-Lipschitz curve. For uniqueness: on an avoidance interval, $d_X(\Phi_t) = d_X(x) + t$ and $d(x, \Phi_t) \leq t$ force $d(x, \Phi_t) = t$ (as $d_X(\Phi_t) \leq d_X(x) + d(x, \Phi_t)$), so $\Phi|_{[0, t]}$ has length $t = d(x, \Phi_t)$ and is a minimising geodesic; the concatenation [foot of x] $\rightarrow x \rightarrow \Phi_t$ has length $d_X(x) + t = d_X(\Phi_t)$, hence minimises, hence has no corner at x : $\Phi|_{[0, t]}$ is the geodesic extension of the minimising geodesic of x , as in the proof of Lemma 5.4(iii). $\square$

Lemma 6.2 (Cut rays: grazing trajectories ride the cut locus to infinity). *Let $z \in \mathcal{A}^*(X) \cap \text{Cut}(X)$. Then the (unique) capture curve Φ from z — the shift of the calibrated ray of z , by Lemma 5.4(iii) — satisfies*

$$\Phi_t(z) \in \mathcal{A}^*(X) \cap \text{Cut}(X), \quad d_X(\Phi_t(z)) = d_X(z) + t, \quad \text{for every } t \geq 0 :$$

an escaping ray that touches the closure of the axis stays in it forever.

Proof. By Lemma 5.4(iii) and its proof, every capture curve from z is the arclength shift of the calibrated ray of z ; it never meets Σ , and the rate-one identity holds for all time. In particular the *canonical* trajectory of §5 is this ray shift. By Lemma 5.1 the cut locus $\text{Cut}(X) = \bar{\Sigma} \cap (M \setminus X)$ is positively invariant under the canonical flow, so $\Phi_t(z) \in \text{Cut}(X)$; and forward points of a calibrated ray lie in $\mathcal{A}^*(X)$ by definition. $\square$

Whether $\mathcal{A}^*(X) \cap \text{Cut}(X)$ can be nonempty for some closed X — merging medial directions accumulating along an escaping calibrated ray — is a regularity question we leave open (it is vacuous for smooth X , by continuity of the cut time; the structure theory of [11, 9] is the natural tool for the general case). The point of Lemma 6.2 is that the theorem below does not need to know: a grazing ray is not an obstruction but a certificate, since it drags medial centres with it to infinity.

Theorem 6.3 (Ideal swallow: the finite layer does not leak). *Let M be complete, X closed, and let $p \in \mathcal{A}^*(X) \cap \mathcal{R}_X^g$ — a swallowed, uncaptured point, class (2) of Theorem 4.4. Then for every finite witness $a_0 \in M_X^g$ with $m_0 := u_p(a_0) > 0$ there is an ideal medial limit function ψ with*

$$\psi(p) \leq -m_0 < 0 :$$

p lies in an ideal free horoball, at depth at least its depth in any finite medial ball.

Proof. Let γ be the calibrated ray of p , $p = \gamma(t_0)$ with $t_0 = d_X(p)$, and let $\eta : [0, L] \rightarrow M$, $L = d(a_0, p)$, be a minimising geodesic from a_0 to p ; it stays in the free ball $B(a_0, d_X(a_0)) \subseteq M \setminus X$. Concatenate:

$$\sigma(s) := \eta(s) \quad (0 \leq s \leq L), \quad \sigma(L+t) := \gamma(t_0+t) \quad (t \geq 0).$$

Along η , $d(\eta(s), p) = L - s$ and $d_X(\eta(s)) \geq d_X(a_0) - s$ give $u_p(\sigma(s)) \geq m_0$; along the ray, $d_X(\gamma(t)) = t$ and $d(\gamma(t), p) = t - t_0$ give $u_p = t_0 \geq m_0$ (the second inequality because $u_p(\sigma(\cdot)) \geq m_0$ persists to $s = L$ by continuity, where $u_p(p) = d_X(p) = t_0$). So

$$u_p(\sigma(s)) \geq m_0 \quad \text{for all } s \geq 0. \quad (3)$$

The endpoint data are opposite: $\sigma(0) = a_0 \in \Sigma$ is captured (at time 0), while $p = \sigma(L) \in \mathcal{A}^*(X)$; the captured set is open and $\mathcal{A}^*(X)$ is closed in $M \setminus X$ (Lemma 5.4), so, using (3),

$$s^* := \inf \sigma^{-1}(\mathcal{A}^*(X)) \in (0, L], \quad x_* := \sigma(s^*) \in \mathcal{A}^*(X), \quad u_p(x_*) \geq m_0,$$

and $\sigma(s)$ is captured for every $s < s^*$.

Case A (the transition grazes the cut locus): suppose there are $s_j \uparrow s^*$ with $\sigma(s_j) \in \bar{\Sigma}$. Then $x_* \in \text{Cut}(X) \cap \mathcal{A}^*(X)$. By Lemma 6.2 the points $w_t := \Phi_t(x_*)$ lie in $\text{Cut}(X)$ with $d_X(w_t) = d_X(x_*) + t \rightarrow \infty$, and by Lemma 6.1 (the trajectory never meets Σ , so $T = \infty$) $u_p(w_t) \geq u_p(x_*) \geq m_0$. Pick $a_t \in \Sigma$ with $d(a_t, w_t) \leq \min(1, 1/t)$: then $a_t \in M_X^g$,

$$u_p(a_t) \geq m_0 - 2/t, \quad d_X(a_t) \geq d_X(x_*) + t - 1/t \rightarrow \infty.$$

Since d_X is bounded on bounded sets, (a_t) has no bounded subsequence. By Lemma 4.2 a subsequence of (ψ_{a_t}) converges locally uniformly to a medial limit function ψ with $\psi(p) = \lim(-u_p(a_t)) \leq -m_0$, and ψ is ideal. Done.

Case B (the transition is clean): otherwise there is $s_1 < s^*$ with $\sigma(s) \notin \bar{\Sigma}$ for $s \in [s_1, s^*)$. Take $s_j \uparrow s^*$ in $[s_1, s^*)$ and set $x_j := \sigma(s_j)$: captured, not in $\bar{\Sigma}$. Each x_j has a capture curve Φ^j with finite first approach T_j , and by the dichotomy proof of Theorem 3.1 there are times $t_j \in [T_j, T_j + 1/j]$ with $b_j := \Phi_{t_j}^j \in \Sigma = M_X^g$. By Lemma 6.1,

$$u_p(b_j) \geq u_p(x_j) - 2/j \geq m_0 - 2/j, \quad d_X(b_j) \geq d_X(x_j) + T_j - 1/j.$$

If (T_j) is unbounded, pass to a subsequence with $T_j \rightarrow \infty$: then $d_X(b_j) \rightarrow \infty$, and exactly as in Case A, Lemma 4.2 produces an ideal ψ with $\psi(p) \leq -m_0$. If (T_j) is bounded, pass to a

subsequence $T_j \rightarrow \bar{T} < \infty$. On $[0, T_j)$ each Φ^j avoids Σ , hence (Lemma 6.1) is the geodesic extension of the minimising geodesic of x_j . Since $x_* \notin \Sigma$ ($\mathcal{A}^*(X) \cap \Sigma = \emptyset$), the direction field U is single-valued and continuous at x_* (Lemma 2.1), so the extensions converge locally uniformly to the extension of the minimising geodesic of x_* — which is the calibrated ray of x_* — and, as $d(b_j, \Phi_{T_j}^j) \leq 1/j$,

$$b_j \longrightarrow z := \Phi_{\bar{T}}(x_*) \in \mathcal{A}^*(X) \cap \bar{\Sigma}, \quad u_p(z) = \lim u_p(b_j) \geq m_0 :$$

a grazing point with full margin. Case A applied at z (its hypothesis is exactly $z \in \mathcal{A}^*(X) \cap \text{Cut}(X)$) finishes the proof. $\square$

Corollary 6.4 (Ideal decomposition of the reconstruction). *For every complete M and closed X ,*

$$\mathcal{R}_X^g = W \cup \bigcup \{ \{ \psi < 0 \} : \psi \text{ an ideal medial limit function of } X \}$$

— the captured set plus the ideal layer; the finite medial balls are needed only where capture already operates. On the Aubry set the trichotomy is decided by the ideal functions alone:

$$\text{class (2)} = \mathcal{A}^*(X) \cap \bigcup_{\psi \text{ ideal}} \{ \psi < 0 \}, \quad \text{class (3)} = \mathcal{A}^*(X) \setminus \bigcup_{\psi \text{ ideal}} \{ \psi < 0 \} :$$

a point escapes unswallowed if and only if it is Aubry and no ideal medial limit function is negative on it. In $\mathbb{R}^n$ this sharpens Theorem 6 of [4]: the half-space layer alone covers the entire drift-swallowed set.

Proof. $W \subseteq \mathcal{R}_X^g$ is Theorem 4.4(1), the horoballs are swallowed by Lemma 4.1, and conversely $\mathcal{R}_X^g \setminus W = \mathcal{A}^*(X) \cap \mathcal{R}_X^g$ is covered by Theorem 6.3. The class identities follow since $\mathcal{A}^*(X) \cap W = \emptyset$. $\square$

Theorem 6.5 (The medial complex carries the reconstruction). *Let M be complete, X closed, and let Ψ be any family of ideal medial limit functions whose free horoballs cover $\mathcal{A}^*(X) \cap \mathcal{R}_X^g$ (by Theorem 6.3 the family of all of them qualifies). Then*

$$\mathcal{U} := \{W\} \cup \{ \{ \psi < 0 \} : \psi \in \Psi \}$$

is an open cover of $\mathcal{R}_X^g$, and the natural map from the homotopy colimit of its Čech diagram (the members and their finite intersections) to $\mathcal{R}_X^g$ is a weak equivalence [6] — a homotopy equivalence, since these spaces are ANRs. Since $M_X^g \hookrightarrow W$ is a homotopy equivalence (Theorem 5.5), the medial complex — M_X^g together with the ideal free horoballs, glued along the pattern of their overlaps with the captured set and with one another — carries the homotopy type of $\mathcal{R}_X^g$; and of $M \setminus X$ whenever the reconstruction is complete.

Proof. Covering is Corollary 6.4 together with $\{ \psi < 0 \} \subseteq \mathcal{R}_X^g$ (Lemma 4.1) and $W \subseteq \mathcal{R}_X^g$; all members are open. The homotopy-colimit statement for an arbitrary open cover is [6]; open subsets of manifolds and their homotopy colimits have CW type, so the weak equivalence upgrades by Whitehead. The last sentence substitutes the CCF equivalence at the captured vertex. $\square$

Corollary 6.6 (Faithfulness without an ideal layer). *If X admits no ideal medial limit function — e.g. whenever M_X^g is bounded, so that no sequence of medial centres escapes — then class (2) is empty, $\mathcal{R}_X^g = W$, and*

$$\mathcal{R}_X^g \simeq M_X^g :$$

the two-geodesic axis carries the homotopy type of everything it reconstructs. This extends the faithfulness regime of Corollary 5.3 beyond bounded distance: d_X may be unbounded and the Aubry set nonempty, as for a round circle in $\mathbb{R}^2$ (axis the centre, $\mathcal{R}_X$ the open disk) or a small circle inside a paraboloid of revolution (axis the vertex, $\mathcal{R}_X^g$ the inner disk, the flaring end escaping) — both with contractible axis and reconstruction.

Proof. Class (2) = $\mathcal{A}^*(X) \cap \bigcup_{\text{ideal}} \{\psi < 0\} = \emptyset$ by Corollary 6.4; so $\mathcal{R}_X^g = W \simeq M_X^g$ by Theorem 5.5. $\square$

Białozyt’s Theorem 1.1 thus generalises to every complete Riemannian manifold: reconstruction by the two-geodesic axis is exactly the union of the sublevel sets of the medial limit functions (Theorem 4.3), total on every compact manifold (Corollary 3.2); the captured part of the complement carries the homotopy type of the axis (Theorem 5.5), totally so when $\sup d_X < \infty$ (Corollary 5.3), and the medial complex carries it for all of $\mathcal{R}_X^g$ unconditionally (Theorem 6.5). The two Euclidean obstructions — the containment step past cut points and the radial offset — are removed, the first by stopping the ascent at its first approach to the axis, the second by normalising the limit objects by their own reach.

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